

Diary of Events, Highlights of Achievements and Memories of the 23rd Indian Antarctic Expedition

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Expedition Leader and Station Commander

- The 23rd Indian Antarctica Expedition Team comprising of 55 expedition members commenced their journey on December 09, 2003 from Dabolim Airport, Goa. to Mumbai and reached Johannesburg on December 10, 2003 (0745 Local Time; 1045 IST).
- The Expedition was formally and ceremonially flagged off at Cape Town in South Africa on December 19, 2003 by Dr. P.C. Pandey, Director, NCAOR, on board MV Emerald Sea. Commenced sailing at 0145 SA time on board MV Emerald Sea.
- The Expedition vessel MV Emerald Sea crossed 60 degrees south Latitude on December 24 and entered the cold waters of South Indian Ocean towards Antarctica. Prepared detailed helicopter flight schedules for deployment of members from each institution based on scientific personnel, equipment and other expedition cargo.
- Crossed 60 degrees south latitude on December 24. Entered 3 m thick pack ice, sailing stopped at 68 degrees south unable to penetrate pack ice. Obtained Ice charts and weather charts from Sanae (SA) and Syowa (Japan) stations in Antarctica. Held meeting with Captain, Chief Officer and Ice Pilot to review steps for resuming sail.
- First contact with Maitri station on 25th December, Christmas Day.

SUMMER PROGRAMMES

With a surge of activities peaking during the crucial daylight months of January and February, a more detailed description of activities and achievements are given below. The normal core programmes and activities follow this for the remaining part of the year. 2004

JANUARY 2004**Logistics**

- With weather gods smiling benevolently, Helicopter sorties commenced promptly on New Year Day on board MV Emerald Sea at 0910 GMT for transporting scientific personnel, logistic team members and scientific cargo. Two helicopters carried out a total of 11 sorties till 2030 GMT shifting 32 expedition members and their cargo to Maitri base station.
- Set foot on Antarctica on January 04 and held day long discussions with Arun Hanchinal, Leader, XXII IAE at Maitri station regarding handing over and taking over.
- Between January 18 and 24 taking over charge of Station Systems was attended to: Boilers, Generators and Vehicles was completed in right earnest. Republic Day celebrations were held at Maitri with guests from Russia, New Zealand, Australia and Poland. This was followed by inauguration of Environmental laboratory. Took overcharge of Maitri station in the Afternoon of January 26.

Scientific Programmes

- a) NGRI recorded first set of Absolute Gravity data on 30th midnight to report value of 982578681.04 Micro- Gals, (b) Daily operation of Seismological Observatory and GPS tracking Station at MAITRI.
- b) GSI Collection of (a) rock samples from Gruber Anorthosite massif for petrological, geochemical and geochronological studies, (b) samples of semi-consolidated clay, silt from moraine terrace for TL dating, (c) Polar ice front movement study using high precision GPS array.
- c) IIG Continuous recording of three component magnetic elements using digital Fluxgate Magnetometer.
- d) NPL Measurements of (a) green house gases using Gas Chromatograph (b) Sun-Photometer MICROTOP for estimating the ozone by water vapor and aerosol optical depth and (c) CO Analyzer.
- e) SOI Crustal movement studies using GPS observations with respect to base station Maitri.

- f) IMD (a) Continuous recording of surface ozone, (b) Vertical profile of ozone by Ozonesonde balloon ascents and (c) Brewer spectrophotometer observations for measurement of total ozone.
- g) NHO Collection of bathymetric and other physical Oceanographic data (sound velocity, sea temperature profile, water conductivity) on board MV Emerald Sea using Acoustic Current Meter and CTD probe.
- h) DEAL Installed Antenna dome (ADE) of INMARSAT-M terminal on the roof of Maitri station and MCU with all accessories (BDE) in Radio room for FAX and E-Mail facilities.

FEBRUARY 2004

- While all the scientific programmes continued from the previous month, the major highlights for this month were (a) carrying out GPS surveys in polar plateau along with magnetic and gravity surveys and (b) sampling of lamprophyre dykes for geochronological dating and terrace sediments for TL-dating.
- On February 28 and 29 all summer camps in and around Maitri station were closed and scientific personnel and equipment shifted back on board MV Emerald Sea.
- Joint Convoy of Army logistics team left by vehicles on February 29.

MARCH 2004

- Preparations for Wintering over begin in the form of a detailed review of Winter Station facilities at Maitri.
- Completed Helicopter reconnaissance of convoy route, shifted to Emerald Sea for unloading operations and flew back to Dakshin Gangotri convoy camp.
- MV Emerald Sea sailed off to Cape Town on March 22, 2004. Returned to Maitri station with first convoy. Wintering Period Commenced.

WINTER PROGRAMME

APRIL 2004

- Glaciological monitoring of Stakes in Dakshin Gangotri Ice shelf during second convoy trip.

- Fire safety and Fire fighting drill, environmental cleanliness and waste management.
- Station Maintenance, Health and Communication facilities.
- Visited Russian Antarctic station, Novolazarevskaya on April 10 for discussions on seismic data requirements and geomagnetic data analog recorder.
- Third convoy returned with fuel and food containers on April 19.

MAY 2004

- Prepared and submitted reports on (a) Scientific programmes and (b) Station equipment.
- Supervised collection of (a) Permafrost data by GSI, (b) Synoptic weather data by IMD and (c) Gas chromatography data of Methane, CO₂ and CO by NPL. Balloon ascents for Ozone measurement launched periodically by IMD.
- Pre-winter stocking of (a) fuel for station generators and boiler systems and (b) food for winter team completed.

JUNE 2004

- The global electric circuit (GEC) equipment for IIG was reinstalled in a new site to improve the quality of data.
- Carried out lake sediment coring for GSI programme at 6.5 m lakebed depth and below 1.2 m frozen surface ice thickness of lake.
- Prepared detailed inventory and demand lists for procurement of 24th IAE pertaining to (a) Clothing items, (b) Food and kitchen items and (c) List of Medicines.
- Mid winter day celebrations on June 21 at Maitri, along with Russian Antarctic winter team members as special invited guests from novolazarevskaya station.
- Received greetings and messages from: (a) Pradhan Mantri Dr. Manmohan singh,

- (b) Raksha Mantri Dr. Pranab Mukherjee, and (c) Vigyan Mantri Shri Kapil Sibal

JULY 2004

- Prepared Demand List for procurement of (a) clothing, (b) Piston Bully vehicles, (c) Summer camp items, (d) Field camp items and (e) Station Complex items for 24th IAE and sent to NCAOR, Goa for procurement.
- Reviewed backup of the digital Seismic data and GPS data by NGRI on CD's for making 2 sets of archival data.

AUGUST 2004

- END OF POLAR NIGHTS : Sunrise at 0930 hours, Sunset at 1300 hours on August 01.
- Reviewed Ozone monitoring programme of IMD with special reference to: (a) continuous surface Ozone measurement, (b) Ozonesonde balloon ascents for vertical Ozone profile and (c) Brewer Spectrophotometer for recording total Ozone.
- INDEPENDENCE DAY CELEBRATED. Flag hoisting outside Maitri station at 1100 hours.
- Reviewed Greenhouse gas monitoring programme of NPL with reference to: (a) Gas chromatograph for measuring Methane, (b) CO Analyzer for measuring atmospheric concentration of Carbon Monoxide and (c) 5-channel measurement of solar radiation and UV by using MICROTOP Sun Photometer, both at Maitri and on Ice shelf.
- Inspected GSI Drilling equipment container on August 29 near Tallaksenvarden Nunatak and HF Wireless REPEATER station at Veteheiya Nunatak on Polar plateau.
- Reconnaissance fieldwork around Long Lake area for planning Lake sediment sampling by drilling.

SEPTEMBER 2004

- On Friday September 03 at 1220 hours, a low flying object resembling a small aircraft with its tail on flame and a trail of smoke was seen for more than thirty seconds. This low flying, burning object was followed

and seen losing height to go down into the ice shelf, half way to horizon, north 55 east of Maitri. This suspected meteorite fall was confirmed by observations from neighboring Russian Antarctic station. Efforts to investigate the area and examine the spot along with convoy team members of Indian Army were not successful.

- Held 2 - day Review Meeting on 05 and 06 September on status of scientific programmes with members of GSI, IIG, IMD, NGRI and NPL.
- Proceeded on CONVOY JOURNEY to India Bay along with Indian Army logistics team for transporting fuel from storage containers to Maitri station. Left Maitri Station at 0650 hours on September 20 and reached India Bay at 2000 hours, 146 km on ice shelf. Supervised decanting of fuel and placement of backloading containers on 21. Carried out monitoring of glaciological stakes for snow ablation and accumulation studies on 22 at Dakshin Gangotri Ice shelf. Left India Bay on September 23 at 1100 hours and reached Maitri station at 2200 hours.

OCTOBER 2004

- On October 15 EME Corps Day was celebrated at Maitri station with all winter team members felicitating EME Logistics Team.
- On October 16 carried out Ice Auger drilling on Long Lake frozen surface for Lakebed sediment sampling. Metal corer broken due to cold fatigue caused to metal components in very low sub zero temperatures.
- On October 26 carried out fieldwork for glaciological studies. Measurement of DG (Dakshin Gangotri) Glacier snout retreat was carried out by GPS Monitoring and tape survey.
- Reviewed work of IIG and NGRI for the polar night period.

NOVEMBER 2004

- On November 01 succeeded in obtaining lake bottom sediment sample in the middle portion of Long Lake (Lat. S 70° 45. 381' & Long. E 11° 39. 585'). Lake sediment coring was resumed using the reducer broken earlier but welded with help from Novo Russian station. The thickness of ice at this site was about 1.65 m and depth of Lake bottom

from the surface was around 5 m. A sediment column of 90 cm was obtained from this point comprising predominantly of dark colored silty clay with sand and algal mats. This sample has been kept in the walk-in-freezer in an upright position and will be used for Biological and Sedimentological studies.

- On November 04, one more hole was drilled through frozen ice surface and 65 cm of sediment column was obtained from lake bottom for TL dating. Due care was taken to ensure that this sample is not exposed to light and the acrylic tube was removed from the corer in a dark room with red bulbs. The water content was siphoned out without disturbing the sediment column and the acrylic tube wrapped with black polythene bags and secured with adhesive tape in the entire length of the acrylic tube after marking the top and bottom and kept for natural dehydration at room temperature. This sample will be used for dating by TL techniques.
- DIWALI CELEBRATED ON NOVEMBER 12 WITH GAIETY AT MAITRI IN SPITE OF A FEELING OF SEPARATION AND LONELINESS AMONG TEAM MEMBERS.
- On November 18, CORPS OF ENGINEERS DAY (SAPPERS DAY) was celebrated at Maitri station with all winter team members felicitating Station Engineers Logistics Team.
- On November 21 last and final convoy left Maitri at 1000 hours. Returned to Maitri base on November 25.

DECEMBER 2004

- On December 03 a multinational 4-member SCAR TEAM - Scientific Committee on Antarctic Research - visited Maitri Station and held discussions and presentations on Antarctic Research programmes. Dr. COLIN SUMMERHAYES, Executive Director, SCAR, Dr. HARTWIG GERNANDT, Head of Antarctic Programme, Germany and Dr. HENRY VALENTINE, Director, South African Antarctic Programme presented the summary and conclusions of recent SCAR meeting regarding main focus on GLOBAL CHANGES IN CLIMATE as priority for all research in Antarctica. I made a PowerPoint presentation of Indian Scientific programme.
- Work on Handing Over Maitri Station to 24th Indian Scientific Expedition to Antarctica has been taken up.

- Reviewed 12 months of work carried out by IMD, IIG, NGRI and GSI for Preliminary Report. Prepared Interim Report on Scientific work carried out.
- Arrangements for Summer Hut accommodation, stocking supplies, stores, additional power and water storage and all other essentials were completed for the New Expedition Team of 54 members expected to arrive from India in the last week of this month. THE XXIV INDIAN SCIENTIFIC ANTARCTIC EXPEDITION TEAM ARRIVED AT MAITRI ON 28.12.2004.

JANUARY 2005

- Supervised and Assisted New Expedition Leader of XXIV ISAE in starting summer team's scientific programmes. Prepared allotment of laboratory space and living accommodation for summer team members in Maitri complex.
- Started preparations for handing over Maitri Station to new Expedition Leader. Detailed list for handing over of items under various heads were prepared.
- HANDED OVER MAITRI STATION TO SHRI RAJESH ASTHANA, EXPEDITION LEADER, XXIV INDIAN SCIENTIFIC ANTARCTIC EXPEDITION ON JANUARY 15 AND SHIFTED ON BOARD SHIP MV EMERALD SEA ALONG WITH ALL WINTER TEAM MEMBERS.
- Supervised helicopter sorties for sending expedition stores and frozen food items to Maitri station from on board ship.
- Supervised and completed operations on Ice shelf for mooring ship, unloading of cargo and decanting of fuel between January 20 and 22.
- PARTICIPATED IN REPUBLIC DAY CELEBRATIONS AT MAITRI ALONG WITH EXPEDITION MEMBERS FROM INDIA, MALAYASIA, AUSTRALIA AND GUESTS FROM RUSSIAN NOVO STATION AND RUSSIAN SHIP CREW.
- Preparations for VIP Team visit were taken up for presenting expedition activities and conducted tour to selected sites in Ice Shelf.

FEBRUARY 2005

- SHRI KAPIL SIBAL, HONOURABLE MINISTER FOR SCIENCE, TECHNOLOGY AND OCEAN DEVELOPMENT VISITED MAITRI STATION AND ARRIVED ON BOARD EXPEDITION VESSEL MV EMERALD SEA AT 1600 HOURS ON FEBRYAURY 03 ALONG WITH EIGHT - MEMBER DELEGATION INCLUDING DOORDARSHAN MEDIA PERSONS.
- Introduced all winter team members to Honourable Minister and held discussions regarding existing facilities and future strategies for Antarctic Scientific Programme. Visited Ice Shelf along with Honourable Minister who inspected Unmanned Magnetic Observatory, Fuel Storage and Cargo Container Complex followed by visit to buried site of Dakshin Gangotri Station and landing on Icebergs. The VIP delegation left for Maitri station on February 04.
- Commenced sailing of ship at 1700 hours on February 08 along with Task Force team for establishing Third Permanent Indian Station in Antarctica at Latitude 70 degrees south; Longitude 76 degrees east - in Larsemann Hills area, Ingrid Christensen Coast, Princess Elizabeth Land, East Antarctica.
- Reached destination point in Larsemann Bay on February 17. Carried out work on placing IGLOO hut, geological studies, sampling of lake waters and GPS surveys. Visited Australian Antarctic Station, DAVIES, Chinese Antarctic Station, ZONGSHAN and Russian Antarctic Station, PROGRESS, in the area adjoining Larsemann Hills.
- Bid farewell to Maitri Station on February 20 and returned to Ship's Bay / India Bay on February 26.

MARCH, 2005

- Supervised and Monitored Joint Convoy operations from Maitri to India Bay. Convoy left Maitri Station on 11th at 0800 GMT and reached India Bay at 0400 GMT on 12th March. Ship MV Emerald Sea started berthing and mooring alongside on Ice Shelf for back loading container cargo arrived by joint convoy from Maitri.
- Completed Back Loading of all container cargo on to Ship MV Emerald Sea. Farewell to Winter team leader and wintering members

of 24th IAE on board ship. Operation of Helicopters closed and both Helicopters dismantled and put down into Ship's cargo hold.

- RETURN VOYAGE OF EXPEDITION VESSEL MV EMERALD SEA COMMENCED FROM ANTARCTICA AT 0600 GMT ON MARCH 13 WITH 56 EXPEDITION MEMBERS ON BOARD FOR RETURNING TO INDIA.
- Expedition Vessel MV Emerald Sea berthed at Cape Town on March 22. Disembarkation of all 56 Expedition Members completed.
- **BACK HOME TO INDIA:** Left Cape Town at 0800 hours on March 26 by SA Airways domestic flight SA – 907152. Reached Johannesburg at 1010 hours. Left Johannesburg at 1210 hours to Mumbai by South African Airways International flight SA – 250. Reached Mumbai at 0015 hours IST on March 27 by South African Airways flight SA – 250.
- Returned to India on completion of Indian Scientific Expedition to Antarctica, as Expedition Leader and Station Commander of XXIII Antarctic Expedition.